

▲

|  |        |
|--|--------|
|  | $\int$ |
|  | $\int$ |

## Finish review

REC-CIS

▲

Finish review

Duration13 days 4 hours

Question 1

Correct

Marked out of 3.00

🚩 Flag question

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and  $153 = 1^3 + 5^3 + 3^3$ .

Example 2:

Input:

REC-CIS

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and  $123 \neq 1^3 + 2^3 + 3^3 = 36$ .

Example 3:

Input:

1634

Output:

true

Note:



🔍 Type here to search



08:23  
13/01/2025



REC-CIS

 $1 \leq N \leq 10^8$ **Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
3
4 int main()
5 {
6     int n;
7     scanf("%d",&n);
8     int x=0,n2= n;
9     while(n2!=0)
10    {
11        x++;
12        n2=n2/10;
13    }
14    int sum=0, n3=n, n4;
15    while(n3!=0)
16    {
17        n4= n3%10;
18        sum=sum+ pow(n4,x);
19        n3=n3/10;
20    }
21    if(n==sum)
22    {
23        printf("true");
24    }
25    else
26    {
27        printf("false");
28    }
29    return 0;
30 }
31
```



Type here to search



08:23

13/01/2025



|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 153   | true     | true  | ✓ |
| ✓ | 123   | false    | false | ✓ |

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints  $1 \leq \text{num} \leq 99999999$   
 Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n,rn,nt=0,i=0;
5      scanf("%d", &n);
6      do
7      {
8          nt=n;
9          rn=0;
10         while(n!=0)
11         {
12             rn=rn*10+n%10;
13             n=n/10;
14         }
15         n= nt+rn;
16         i++;
17     }while(rn!=nt || i==1);
18
19     printf("%d",rn);
20     return 0;
21 }
```

REC-CIS

```
20 | return 0;
21 | }
```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 32    | 55       | 55    | ✓ |
| ✓ | 789   | 66066    | 66066 | ✓ |

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

REC-CIS

33

Explanation:

Here the lucky numbers are 3, 4, 33, 34., and the 3rd lucky number is 33.

Sample Input 2:

34

Sample Output 2:

33344

**Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n=1, i=0, nt, co=0, e;
5     scanf("%d", &e);
6     while(e>0)
```



🔍 Type here to search

08:23  
13/01/2025

2

REC-CIS

```

5 scanf("%d", &e);
6 while(i<e)
7 {
8     nt=n;
9     while( nt!= 0)
10 {
11         co=0;
12         if(nt %10 !=3 && nt%10!=4)
13 {
14             co=1;
15             break;
16         }
17         nt= nt/10;
18     }
19     if(co==0)
20 {
21         i++;
22     }
23     n++;
24 }
25 printf("%d",--n);
26 return 0;
27 }

```

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 34    | 33344    | 33344 | ✓ |

Passed all tests! ✓

Finish review



🔍 Type here to search



08:23

13/01/2025





**Duration** 26 days / hours

Question 1  
Correct  
Marked out of  
3.00  
[Flag question](#)

Input format:

The lines after that contain a different values for size of the chessboard

Print a chessboard of dimensions size \* size. Print a W for white spaces and B for black spaces.

2  
3  
5

WBW  
BWB

REC-CIS

WBW  
WBWBW  
BWBWB  
WBWBW  
BWBWB  
WBWBW

**Answer:** (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,size;
5     scanf("%d",&T);
6     for (int t=0;t<T;t++){
7         scanf("%d",&size);
8         for (int i=0;i<size;i++) {
9             for (int j=0;j<size;j++) {
10                if((i+j)%2==0) {
11                    printf("W");}
12                else {printf("B");}
13            }
14            printf("\n");
15        }
16    }
17    return 0;
18 }
19
20
```



Type here to search



08:22  
13/01/2025



REC-CIS

|   | Input | Expected | Got   |   |
|---|-------|----------|-------|---|
| ✓ | 2     | WBW      | WBW   | ✓ |
|   | 3     | BWB      | BWB   |   |
|   | 5     | WBW      | WBW   |   |
|   |       | WBWBW    | WBWBW |   |
|   |       | BWBWB    | BWBWB |   |
|   |       | WBWBW    | WBWBW |   |
|   |       | BWBWB    | BWBWB |   |
|   |       | WBWBW    | WBWBW |   |

Passed all tests! ✓

Question 2  
Correct  
Marked out of 5.00  
[Flag question](#)

Let's print a chessboard!

Write a program that takes input:

The first line contains T, the number of test cases

Each test case contains an integer N and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

REC-CIS

Print the chessboard as per the given examples

Sample Input / Output

Input:

2

2 W

3 B

Output:

WB

BW

BWB

WBW

BWB

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,N;
5     char c;
6     scanf("%d" &T);
```



1



10

REC-CIS

Question 3

Correct

Marked out of 7.00

[Flag question](#)

Decode the logic and print the Pattern that corresponds to given input.

If N= 3

then pattern will be :

10203010011012

\*\*4050809

\*\*\*\*607

If N= 4, then pattern will be:

1020304017018019020

\*\*50607014015016

\*\*\*\*809012013

\*\*\*\*\*10011

Constraints

$2 \leq N \leq 100$



Type here to search



08:22  
13/01/2025





1



1

REC-CIS

```
*****
**4050809
****607
Case #2
1020304017018019020
**50607014015016
****809012013
*****10011
Case #3
102030405026027028029030
**6070809022023024025
****10011012019020021
*****13014017018
*****15016
```

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,v,p3,c,in,i,i1,i2,t,ti;
5     scanf("%d",&t);
6
7     for(ti=0;ti<t;ti++)
8     {
9         v=0;
10        scanf("%d",&n);
11        printf("Case #%d\n",ti+1);
```



## REC-CIS

```
5 scanf("%d",&t);
6
7 for(ti=0;ti<t;ti++)
8 {
9     v=0;
10    scanf("%d",&n);
11    printf("Case #%d\n",ti+1);
12    for(i=0;i<n;i++)
13    {
14        c=0;
15        if(i>0)
16        {
17            for(i1=0;i1<i;i1++)
18                printf("***");
19        }
20        for(i1=i;i1<n;i1++)
21        {
22            if(i>0)
23                c++;
24            printf("%d0",++v);
25        }
26        if(i==0)
27        {
28            p3=v+(v*(v-1)+1);
29            in=p3;
30        }
31        in=in-c;
32        p3=in;
33        for(i2=i;i2<n;i2++)
34        {
35            printf("%d",p3++);
36            if(i2!=n-1)
37                printf("0");
38        }
39        printf("\n");
40    }
41 }
42 }
```